blood pressure: random-effects meta-analysis (standardized mean difference)

Comparison: Intervention / Control

Methods

Standardized mean differences were pooled using a random-effects model with the DerSimonian–Laird estimator of between-study variance. Confidence intervals (95%) used the normal approximation. Heterogeneity was assessed with Cochran's Q and I^2. Analyses were performed in RC MetaStudio 0.5.1. R packages loaded for this analysis: metafor 5.0.1, meta 8.5-0.

Results

6 studies were included (500 participants in Intervention, 495 in Control). The pooled standardized mean difference was 0.358 (95% CI [0.152, 0.565], z = 3.400, p < .001). The confidence interval excludes 0, with higher mean values in Intervention than in Control. Heterogeneity: Q = 11.914, df = 5, p = .036. I^2 = 58.0%, which may represent moderate to substantial heterogeneity (Cochrane Handbook ranges). Between-study variance was τ^2 = 0.037. The 95% prediction interval was -0.072 to 0.789.

Table 1. Pooled effect

Effect	Estimate	95% CI	SE	z	p
Standardized mean difference	0.358	0.152 to 0.565	0.105	3.400	< .001

Table 2. Model fit

Statistic	Value
Studies (k)	6
Participants in Intervention	500
Participants in Control	495
Cochran’s Q	11.914
Heterogeneity degrees of freedom	5
Heterogeneity p	.036
I^2	58.0%
Between-study variance (τ^2)	0.037
95% prediction interval	-0.072 to 0.789

Table 3. Study results

Study	Effect	95% CI	Weight
Carroll	0.095	-0.264 to 0.453	15.8%
Grant	0.277	-0.068 to 0.623	16.3%
Peck	0.367	-0.075 to 0.808	12.6%
Donat	0.664	0.463 to 0.866	23.3%
Stewart	0.462	0.054 to 0.870	13.8%
Young	0.185	-0.116 to 0.486	18.3%

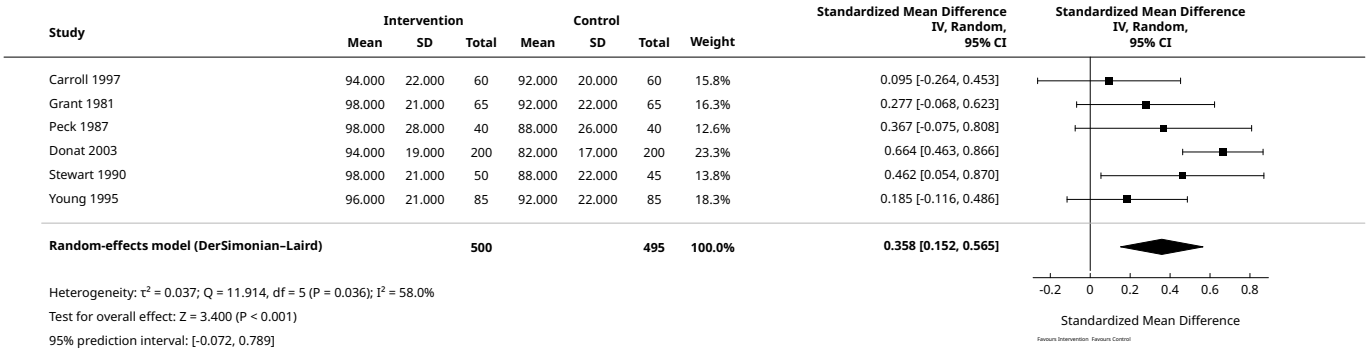

Figure 1. Forest Plot

Supplementary details

Table 4. Full-precision results

Source	Results
Summary — Model Results	Estimate: 0.358 Lower bound (95% CI): 0.152 Upper bound (95% CI): 0.565 SE (standardized scale): 0.105 p-value: < 0.001
Summary — Heterogeneity	Q: 11.914 df: 5 p-value: 0.036 I^2: 58.0% τ^2: 0.037
Summary — Test for overall effect	Z: 3.400 p-value: < 0.001
Summary — Prediction interval	Lower bound (95% PI): -0.072 Upper bound (95% PI): 0.789
Weights — Study weights	Study: Carroll Effect: 0.095 Lower bound: -0.264 Upper bound: 0.453 Weight (%): 15.8%
Weights — Study weights	Study: Grant Effect: 0.277 Lower bound: -0.068 Upper bound: 0.623 Weight (%): 16.3%
Weights — Study weights	Study: Peck Effect: 0.367 Lower bound: -0.075 Upper bound: 0.808 Weight (%): 12.6%
Weights — Study weights	Study: Donat Effect: 0.664 Lower bound: 0.463 Upper bound: 0.866 Weight (%): 23.3%
Weights — Study weights	Study: Stewart Effect: 0.462 Lower bound: 0.054 Upper bound: 0.870 Weight (%): 13.8%
Weights — Study weights	Study: Young Effect: 0.185 Lower bound: -0.116 Upper bound: 0.486 Weight (%): 18.3%

Pooled Standardized Mean Difference: 0.358 (95% CI 0.152 to 0.565).

95% prediction interval: the underlying effect in a new study; this is not the confidence interval of the pooled estimate.

References

1. Random-effects meta-analysis: DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. Controlled Clinical Trials, 7(3), 177-188. doi:10.1016/0197-2456(86)90046-2.

2. Random-effects meta-analysis: Hedges, L. V. (1983). A random effects model for effect sizes. Psychological Bulletin, 93(2), 388-395. doi:10.1037/0033-2909.93.2.388.

3. Implementation reference: Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. Journal of Statistical Software, 36(3), 1-48. doi:10.18637/jss.v036.i03.